CLC^{2}SIE: Cross-Level Consistency Constraint for Semisupervised Building Instance Extraction From High-Resolution Remote Sensing Imagery

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***Abstract*— Semisupervised building instance extraction aims to learn from limited labeled data alongside an extensive collection of unlabeled data, offering a promising approach for extracting building instances from high-resolution (HR) remote sensing imagery (RSI). However, complex RSIs frequently encounter challenges such as background interference and intricate noise, struggling in generating reliable pseudo labels. To alleviate this, we propose a novel cross-level consistency constraint semisupervised building instance extraction (CLC^{2}SIE) method to enhance pseudo label generation. Specifically, CLC^{2}SIE contains two core modules: object-level dynamic consistency (OLDC) and pixel-level saliency consistency (PLSC). The OLDC module dynamically converts building features from background into valuable supplementary information, enhancing the model's perception of building instances in complex scenes. Additionally, the PLSC module is designed to mitigate the boundary noise in pseudo labels by saliency guidance, which improves model's awareness of building contours. By colearning these modules in an end-to-end manner, CLC^{2}SIE facilitates pseudo label generation and improves extraction performance. Experiments were conducted on the three public building datasets, i.e., WHU, CrowdAI and TCC, which demonstrate that CLC^{2}SIE achieves superior performance compared to state-of-the-art semisupervised instance extraction methods at different labeling ratios. This study explores a novel semisupervised learning (SSL) framework that exploits cross-level consistency to improve pseudo label generation, offering a methodological reference for various SSL applications in RSIs.**

***Index Terms*— Building instance extraction, consistency constraint, high-resolution (HR) remote sensing imagery (RSI), semisupervised learning (SSL).**

I. Introduction

As a foundational task of remote sensing imagery (RSI) processing, building instance extraction extensively serves for urban planning and construction, environmental monitoring and management, disaster preparedness and mitigation, and map updating [1], [2], [3], [4]. With the rapid development of satellite remote sensing technology, a large number of high-resolution (HR) RSIs have been continuously produced, providing valuable resources for building extraction. However, the intricate designs and varied scales of buildings, along with natural elements such as trees and shadows, bring additional complexity [5]. As a consequence, achieving precise and automatic extraction of building instances from HR RSIs still poses significant challenges.

Traditional extraction methods manually select low-level discriminative semantic features from RSIs to identify buildings, such as edge [6], spectrum [7], color [8], or textures [9]. Usually, these features highly rely on extensive expert experience and manually crafted rules, exhibiting limited robustness in complex and varied environments. Recently, with the rapid advancement of deep learning technology, deep feature-based instance segmentation methods have delivered impressive results on building instance extraction task [10], [11], [12], [13]. These methods automatically learn and interpret abstract high-dimensional features from low-dimensional features, and extract buildings in a first-detect-then-segmentation manner. However, the reliance on pixel-level annotations for these methods constitutes a considerable limitation, as obtaining such intricate annotations is frequently labor-intensive and impractical for practical applications [14].

In recent years, semisupervised learning (SSL) has garnered attention as a sort of approach to mitigate the reliance on data annotations [15], [16]. It allows to learn from a limited quantity of labeled data alongside an extensive collection of unlabeled data. Among existing SSL building instance extraction works, the pseudo-labeling approach based on confidence has attained remarkable performance [17]. In this paradigm, pseudo labels are assigned to unlabeled data after filtering by a preset threshold. Then, both the labeled RSIs and pseudo-labeled RSIs are used to train a semisupervised model. In practice, these methods still face two challenges. First, RSIs encompass ground objects that vary widely in scale and type, introducing complex background interference during pseudo label generation. Second, the performance of these confidence-based methods relies heavily on the quality of pseudo labels [18]. When a substantial number of noisy pseudo

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labels are present, the model suffers from fit bias, resulting in performance degradation [19].

To address these limitations, the present study proposes a novel SSL building instance extraction method that implements pseudo label optimization in accordance with cross-level consistency constraints, termed as CLC^{2}SIE. The method includes two core modules: object-level dynamic consistency (OLDC) and pixel-level saliency consistency (PLSC). First, the OLDC module constrains the consistency of object box level, dynamically enhancing the weights of reliable bounding boxes while suppressing unreliable ones. This converts background interference into valuable supplementary information during the detection stage, enhancing the model's ability to perceive building instances in complex scenes. Subsequently, the PLSC module provides a more reliable guidance by saliency boundaries for pixel-level mask consistency constraint, mitigating the noise in pseudo labels and improving model's awareness of building contours. By properly training these two components jointly in an end-to-end manner, CLC^{2}SIE can improve the quality of pseudo label and boost the performance of SSL building instance extraction. The main contributions of this study can be summarized as follows.

1) This study proposes an end-to-end semisupervised framework for building instance extraction, called CLC^{2}SIE, which iteratively promotes pseudo label generation by cross-level consistent constraint.
2) The OLDC module dynamically converts building features from complex background to better perceive and distinguish building instances. The PLSC module mitigates the boundary noise in pseudo labels by saliency awareness guidance. By colearning these modules enables the proposed semisupervised framework to achieve the promotion of pseudo label and building instance extraction.
3) Experiments on three datasets demonstrate that the proposed method considerably outperforms the full-supervised and semisupervised instance extraction methods, with notable improvements in the quality of pseudo labels.

The remainder of this article is organized as follows. Section II provides a brief review of related works. Section III describes the details of the proposed method. In Section IV, we reported and analyzed the experimental results obtained. Section V discusses ablation studies and some factors that may impact model performance. Finally, Section VI presents the conclusions drawn from this study.

II. Related Work

Sections II-A and II-B present the most relevant works in the field of study, including building extraction from RSIs and SSL building instance extraction.

A. Building Extraction From RSIs

Building extraction is an extremely significant task in the field of remote sensing. Previous methods of building extraction from RSIs can be broadly divided into two groups: manual feature-based approaches and deep learning-based approaches.

Traditional manual feature-based methods focus on the low-level feature descriptors designed by experts, such as edge, spectral, shadow, and color [20]. For instance, Chen et al. [6] employed edge regularity indices and shadow line indices to achieve satisfactory performance in delineating building regions. Xu et al. [7] integrated four correlative RGB-D priors for improved rooftop extraction to address the complex issues of varying appearance and large-scale range. Sirmacek and Unsalan [8] proposed that improved detector uses texture information from both LIDAR and ortho imagery to achieve more effective separation of buildings from trees. While traditional methods have achieved decent performance, their reliance on expert knowledge and lack of high-level semantic feature descriptions limit their potential in complex RSIs.

Recently, deep learning-based methods have gained significant attention due to the impressive feature representation capabilities of convolutional neural networks (CNNs) [21]. Many researchers utilize semantic segmentation to facilitate automatic building extraction, in which buildings are extracted by classifying RSI pixels as buildings or nonbuildings. For example, Zhu et al. [22] proposed E-D-Net based on FCN [23], which integrates edge information with fine details to alleviate the poor boundaries problem to some extent. Li et al. [24] introduced spatial and channel attention mechanism and multiscale fusion module to enhance U-Net [25] for building extraction. Although these methods have advanced building extraction, they only can classify an RSI into building and nonbuilding areas. Namely, they cannot extract individual building instances.

In recent years, advanced approaches derived from instance segmentation models have been explored to extract building instances from RSIs, yielding promising results [26]. Represented by Mask R-CNN [27], Zhang et al. [10] developed DF-Mask head into Mask R-CNN, which improves the mask features through the gated directional context-aware modules and a direction field rectification mechanism. Fang et al. [28] introduced an attention-based feature pyramid subnetwork and a contour subnetwork to improve the instance extraction performance of small-scale buildings. Liu et al. [29] proposed a lightweight two-stage instance extraction method capable of adaptively optimizing the mask contour by the guided anchoring and iterative subdivision. Furthermore, recent advancements in foundational models, such as segment anything model (SAM) [30], can facilitate the building extraction from RSIs by efficient fine-tuning and training on extensive labeled datasets. Luo et al. [31] introduced adapters to fine-tune backbone and mask decoder of SAM progressively, and extracted targets from RSIs by multistage sequential training. The effectiveness of these methods hinges on access to extensive pixel-level annotation data. Nevertheless, manually labeling a substantial volume of RSIs is a labor-intensive endeavor, which poses a significant barrier to practical application.

B. SSL Building Instance Extraction

SSL is an effective paradigm for alleviating data annotation dependencies, which improves performance by using limited

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labeled data and a large amount of unlabeled data [32]. Recent advances have proposed semisupervised instance segmentation (SSIS) for individual object segmentation in natural scene images [33], [34], [35], [36], [37], [38]. They inherit the pseudo-labeling idea of SSL that generates instance-level pseudo labels for unlabeled data to participate in training. For example, noisy boundary (NB) [33] was the first to formally propose the SSIS task, which designs noise-tolerant and boundary-preserving strategies to alleviate the negative effects brought by noisy pseudo labels. Rumberger et al. [34] proposed an augmentation consistency training method for instance segmentation, which combines lose-masking scheme and momentum-updated teacher to improving data efficiency in SSIS. Wu et al. [36] combined the parallel instance segmentation model with the instance consistency regularization framework to generate and utilize high-quality instance pseudo labels. Hu et al. [35] designed the DALoss that aligns the classification loss and the segmentation loss term based on the quality of pseudo labels. However, all these methods initially designed for natural image containing a main object, may not guarantee a superior advantage in extracting buildings from RSIs with multiple instances.

For remote sensing field, Fang et al. [17] proposed SSBIE, the first SSL building instance extraction method, which integrates teacher–student learning and pseudo labeling to improve the building instance extraction. SSBIE is a typical pseudo-labeling method, which uses the adaptive threshold to filter pseudo labels with low confidence. Despite achieving impressive outcomes, this method disregards the multiple background interference in RSIs. Consequently, it is prone to producing unreliable pseudo labels for unlabeled RSI, which ultimately degrades the extraction performance.

III. Methods

A. Overview of CLC^{2}SIE

The goal of CLC^{2}SIE is to improve the generation of pseudo labels by cross-level consistency constraint, and better extract building instances from HR RSIs. Given a set of RSIs, the labeled data is denoted as $D_s = \{x_s^i, \mathcal{G}^i\}_{i=1}^{N_s}$, where x_i^s represents the labeled data, $\mathcal{G}^i$ denotes the corresponding label, and N_s is the number of D_s. Similarly, the unlabeled data is denoted as $D_u = \{x_u^i\}_{i=1}^{N_u}$, where x_u^i represents the unlabeled data, N_u is the number of D_u, and $N_u \gg N_s$.

The proposed CLC^{2}SIE is illustrated in Fig. 1. CLC^{2}SIE is an end-to-end SSL building instance extraction method, which employs the teacher–student cotraining architecture. In the training stage, the parameters of the teacher model and the student model will be randomly initialized. The student model updates its parameters by backpropagation, while the teacher model is updated by the exponential moving average (EMA) [39] of the student model's parameters.

In each training iteration, labeled and unlabeled RSIs are randomly sampled with the predefined ratio to form a data batch. Then, they are fed into the subsequent teacher–student model for training. For the labeled data, the student model is trained with the ground-truth labels. For the unlabeled data, consistency regularization is introduced to constrain consistency across various perturbed versions of the origin image [40]. Specifically, "weak" data augmentation (WDA) and "strong" data augmentation (SDA) are employed to obtain different views of unlabeled RSIs, respectively. The "weak" view is employed to generate pseudo labels for buildings in the teacher model. The "strong" view is utilized to derive predictions for the same buildings in the student model, thereby enforcing consistency constraints with these pseudo labels. The SSL building instance extraction task follows the manner of first detect then segment, so the consistency constraint is defined as the combination of classification loss and regression loss in the detection level, and mask loss in the segmentation level.

Specifically, CLC^{2}SIE enforces consistency constraints through two core modules: the OLDC module and the PLSC module. In the OLDC module, by dynamically adjusting the weights of candidate boxes, background interference is transformed into valuable supplementary information, enhancing the model's perception of building instances in complex scenes. Subsequently, the PLSC module improves model's sensitivity of building contours by saliency awareness guidance. These two processes effectively address the challenges posed by interference from complex background in RSIs and boundary noise in pseudo labels, respectively. Each module of the proposed method will be elaborated in the following.

B. Object-Level Dynamic Consistency

The OLDC module is designed to leverage pseudo labels predicted by the teacher model as training targets, which in turn guide the student model in learning the mappings from features to categories and locations. In general, the student model utilizes the classification scores associated with the "building" category to select candidate bounding boxes. Following prior work [33], these candidate boxes B are partitioned into foreground b_{fg} and background b_{bg} sets by a specified threshold, as expressed below

$$\begin{aligned} b_{\text{fg}} &= \left\{ B^i \mid p^i \geq \tau, i = 1, 2, \ldots, N_{\text{fg}} \right\} \\ b_{\text{bg}} &= \left\{ B^j \mid p^j < \tau, j = 1, 2, \ldots, N_{\text{bg}} \right\} \end{aligned} \tag{1}$$

where τ is the confidence threshold, and $p \in [0, 1]$ denotes the classification score of a candidate bounding box for the "building" category. N_{fg} and N_{bg} are the number of foreground and background bounding boxes, respectively. Candidate bounding boxes classified as background are directly discarded, as its low classification probability indicates minimal likelihood of containing building instances. However, RSI scenarios are intricate and frequently contain interfering objects that resemble buildings, such as shipping containers, trees, and shadows. Taking a low threshold to select candidate boxes may lead to the incorrect inclusion of these interfering objects as foreground boxes, as illustrated by the circled area in Fig. 2(a). In contrast, applying a high threshold can filter out low-confidence predictions, but may also erroneously exclude true building-containing boxes as negative samples, as shown in the red rectangles of Fig. 2(b) and (c). Such mislabeling can adversely affect the training process and ultimately degrade the performance of building instance extraction.

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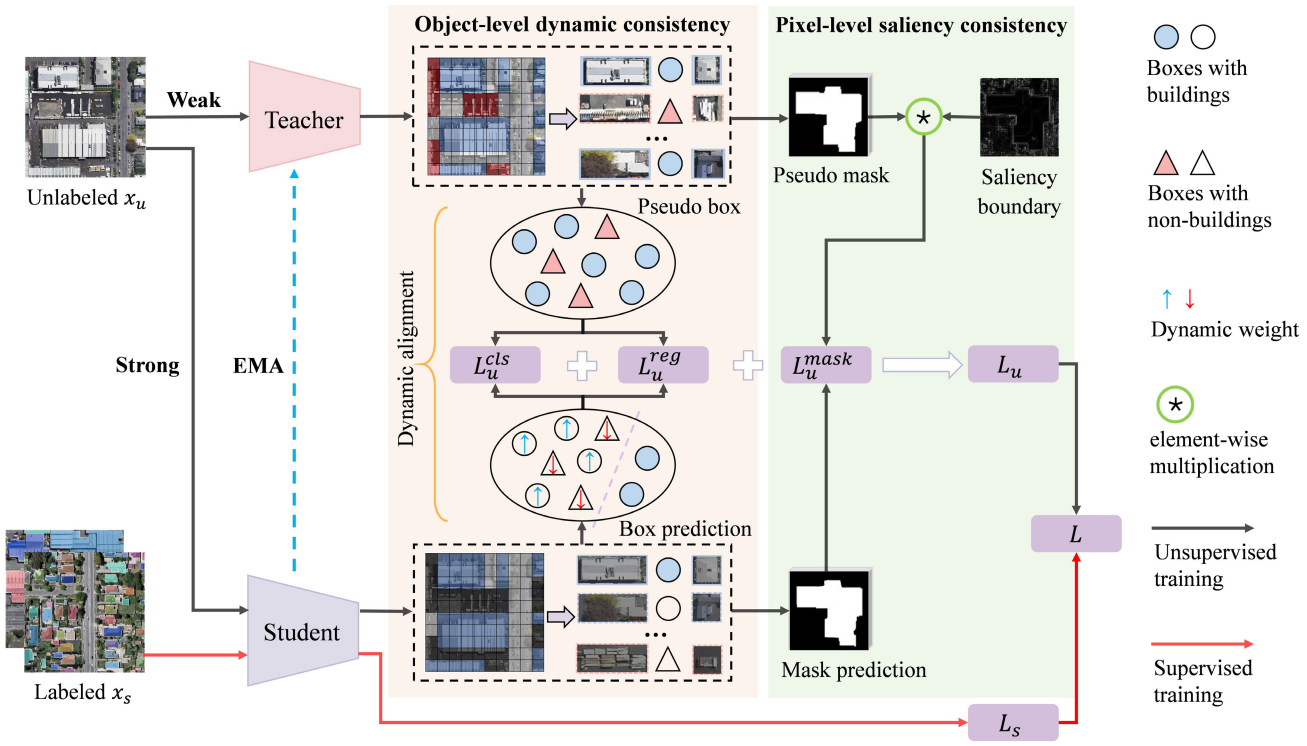


Fig. 1. Overview of the CLC^{2}SIE. First, labeled and unlabeled RSIs are randomly sampled to form a batch for training. The labeled data is used to calculate the supervision loss by real labels, and the unlabeled data is fed into the OLDC module after strong and weak data augmentation. Then the OLDC module constrains the consistency of pseudo boxes and prediction boxes by dynamic alignment strategy, and calculates unsupervised classification and regression loss in the detection level. Subsequently, the PLSC module provides a more reliable guidance by saliency boundary for pixel-level consistency constraint with unsupervised mask loss. Finally, by combining the supervised losses and unsupervised losses, the teacher and student models are progressively trained to generate reliable pseudo labels.



Fig. 2. Example of candidate boxes filtered by different thresholds. Note: the orange circles highlight interfering objects misidentified as foreground, while red rectangles mark true buildings erroneously filtered out as background. (a) Threshold 0.2. (b) Threshold 0.5. (c) Threshold 0.8.

To alleviate this issue, we design the dynamic alignment strategy into the OLDC module that exploits the richer information in the candidate boxes, as shown in Fig. 3(b). Specifically, the probability of each candidate box generated by the student model as a building is first evaluated, and then they are used to weight loss. According to the threshold, a set of candidate boxes is divided into foreground and background, the classification loss of unlabeled data is defined as follows:

$$L_u^{\text{cls}} = \frac{1}{N_{\text{fg}}} \sum_i^{N_{\text{fg}}} l_{\text{cls}}\left(b_{\text{fg}}^i, \mathcal{R}_{\text{cls}}\right) + \frac{1}{N_{\text{bg}}} \sum_j^{N_{\text{bg}}} \left(\lambda^j \cdot l_{\text{cls}}\left(b_{\text{bg}}^j, \mathcal{R}_{\text{cls}}\right)\right) \tag{2}$$

where $\mathcal{R}_{\text{cls}}$ represents the probability score of "building" category in pseudo labels generated by the teacher model. $l_{\text{cls}}(\cdot)$ denotes the classification loss, typically the cross-entropy loss. The dynamic weight λ^j is formulated as follows:

$$\lambda^j = \frac{p^j}{\sum_k^{N_{\text{bg}}} p^k}. \tag{3}$$

In (2), the dynamic weight assignment can be interpreted as an instance of the attention mechanism, greater weights

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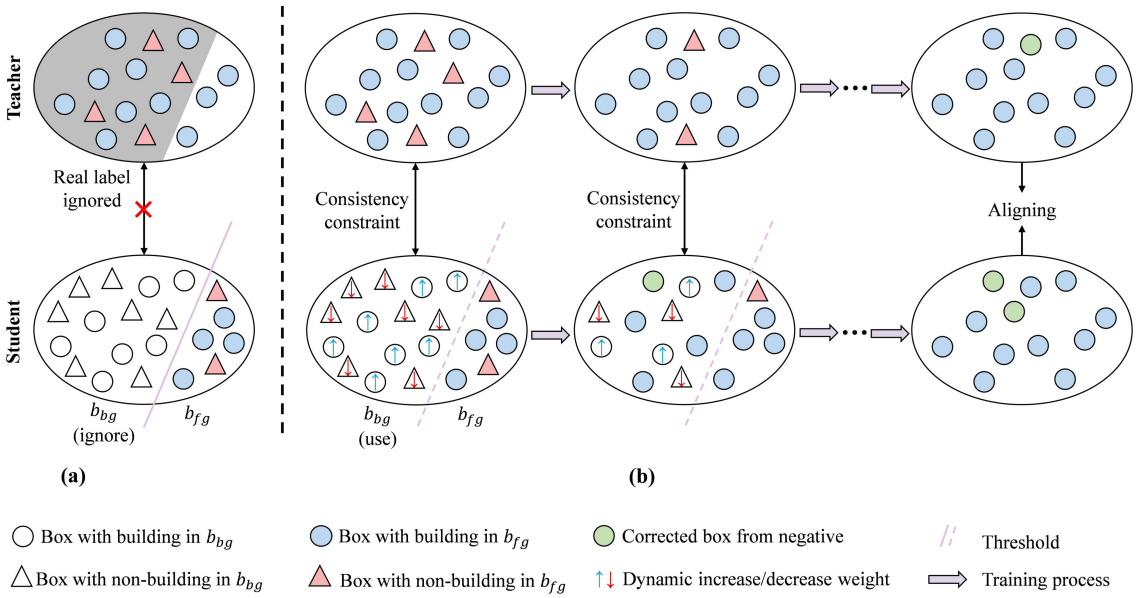


Fig. 3. (a) Comparison between threshold filtering strategy and (b) our dynamic alignment strategy. The threshold filtering strategy neglects b_{bg} from student model, leading to boxes with buildings being disregarded during consistency training. Unlike the former, our strategy transforms boxes with buildings in b_{bg} into valuable supplementary information and dynamically adjusts their weights to achieve alignment. Moreover, potential building-containing boxes that may have been misclassified as background are reevaluated and assigned appropriate weights, improving the model's perception of buildings during training.

are allocated to the boxes containing building with higher confidence, akin to the self-attention strategy [41]. Additionally, nonbuilding interferences with lower confidence receive smaller weights, ensuring that uncertain predictions contribute less to the optimization process, thereby enhancing the robustness of semisupervised training [42]. This converts valuable information from background to enhance the model's ability to perceive building instances in complex scenes. Note that in the absence of foreground bounding boxes, the detrimental effects induced by background bounding boxes are exacerbated by λ, which will break from this training iteration to the next.

For box regression, the bounding boxes in the foreground that have the largest IOU with the pseudo labels are calculated, which is defined as follows:

$$\widetilde{b}_{\text{fg}} = \arg\max_{b^i_{\text{fg}}} \left(\left\{ \ell_{\text{IOU}} \left(b^i_{\text{fg}}, \mathcal{R}_{\text{reg}} \right) | i \epsilon N_{\text{fg}} \right\} \right) \tag{4}$$

where $\mathcal{R}_{\text{reg}}$ represents the pseudo boxes generated by the teacher model for regression. Therefore, the regression loss of unlabeled data can be defined as follows:

$$L_u^{\text{reg}} = \frac{1}{N_{\text{fg}}} \sum_i^{N_{\text{fg}}} l_{\text{reg}} \left(\widetilde{b}^i_{\text{fg}}, \mathcal{R}_{\text{reg}} \right) \tag{5}$$

where $l_{\text{reg}}(\cdot)$ represents the regression loss of the bounding box.

C. Pixel-Level Saliency Consistency

After the OLDC module, CLC²SIE further implements the consistency between the pseudo mask of the teacher model and the predicted mask of the student model via PLSC module. However, noise inherently exists in the pseudo masks generated by the teacher model. These noises primarily arise near building contours, where pixels lie along decision boundaries and discriminative features are not prominent [17]. In this case, if the noisy pseudo labels are directly guided to the student model, as a general supervised training does when real mask annotations are provided, the wrong signals will be amplified during training, deteriorating the performance.

To mitigate the problem, we propose saliency awareness to provide a more reliable guidance through the pseudo labels weighted with saliency boundaries. Specifically, the saliency boundaries are extracted from the buildings of RSIs, where Laplace [43] operator is adopted. The resulting binary mask t_u^i can be expressed as follows:

$$t_u^i = \Gamma \left(x_u^i \right) \tag{6}$$

where $\Gamma(\cdot)$ represents the boundary extraction operation. Taking t_u^i as a guided weight, the consistency mask loss L_u^{mask} can be expressed as follows:

$$L_u^{\text{mask}} = \frac{1}{N_u^{\text{mask}}} \sum_i^{N_u^{\text{mask}}} l_{\text{mask}} \left(\hat{y}^i_{\text{mask}}, t_u^i \cdot \mathcal{R}_{\text{mask}} \right) \tag{7}$$

where N_u^{mask} denotes the number of building masks, $\hat{y}_{\text{mask}}$ is the prediction mask of the student model, and $\mathcal{R}_{\text{mask}}$ represents the pseudo masks generated by the teacher model. $l_{\text{mask}}(\cdot)$ represents the building mask loss, typically the binary cross-entropy loss.

As shown in Fig. 4, weighting masks of pseudo labels by saliency boundaries can highlight the contours of buildings. In contrast, the masks predicted by the student model from unlabeled RSIs have poor perception ability for building contour. By constraining the mask difference of the two outputs to be consistent during training, the noises in the pseudo labels are mitigated and the model's sensitivity of building contours is enhanced.

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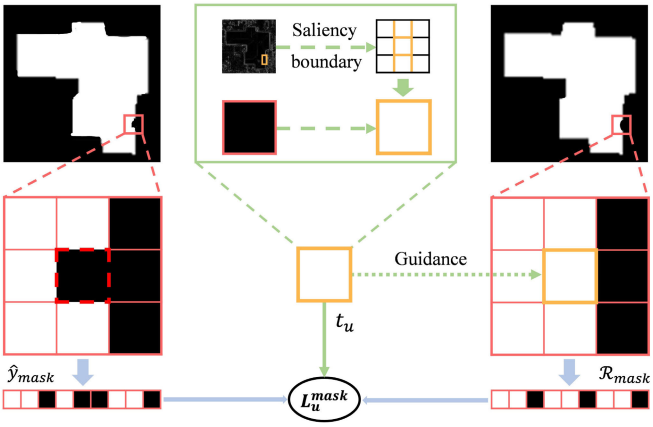


Fig. 4. Graphical illustration of the saliency awareness. Taking the saliency boundary t_u from the RSI to mitigate the noise in the pseudo mask $\mathcal{R}_{\text{mask}}$ and highlight the boundary pixels. Then, the prediction mask $\hat{y}_{\text{mask}}$ from the student model is constrained by consistency loss, thereby enhancing the model's awareness of building contours.

D. Cross-Level SSL Joint Loss Function

The loss of student model training mainly consists of two parts, including supervised loss and unsupervised loss. The supervised loss L_s is calculated by the labeled data with ground truth for building class loss, regression loss, and mask loss. L_s is defined as follows:

$$L_s = \frac{1}{N_s}\sum_{i}^{N_s}\left(L_s^{\text{cls}}\left(y_{\text{cls}}^i, \mathcal{G}_{\text{cls}}^i\right) + L_s^{\text{reg}}\left(y_{\text{reg}}^i, \mathcal{G}_{\text{reg}}^i\right) + L_s^{\text{mask}}\left(y_{\text{mask}}^i, \mathcal{G}_{\text{mask}}^i\right)\right) \tag{8}$$

where y_{cls}^i, y_{reg}^i, and y_{mask}^i represent the categories, bounding boxes, and masks of building instances predicted by the student model on the labeled data.

The unsupervised loss is calculated based on the consistency between the student model's predictions on unlabeled RSIs and the pseudo labels generated by the teacher model. It includes the classification loss and regression loss at the detection level, and the mask loss at the segmentation level. By substituting (1), (4), and (6), L_u can be expressed as follows:

$$L_u = \frac{1}{N_u}\sum_{i}^{N_u}\left(L_u^{\text{cls}}\left(\hat{y}_{\text{cls}}^i, \mathcal{R}_{\text{cls}}^i\right) + L_u^{\text{reg}}\left(\hat{y}_{\text{reg}}^i, \mathcal{R}_{\text{reg}}^i\right) + L_u^{\text{mask}}\left(\hat{y}_{\text{mask}}^i, \mathcal{R}_{\text{mask}}^i\right)\right) \tag{9}$$

where $\hat{y}_{\text{cls}}^i$, $\hat{y}_{\text{reg}}^i$, and $\hat{y}_{\text{mask}}^i$ represent the categories, bounding boxes, and masks of building instances predicted by the student model on the unlabeled data. $\mathcal{R}_{\text{cls}}^i$, $\mathcal{R}_{\text{reg}}^i$, and $\mathcal{R}_{\text{mask}}^i$ are the pseudo labels predicted by the teacher model on unlabeled data, including categories, bounding boxes, and masks of building instances.

By combining (7) and (8), the total loss of the student model is given by

$$L = L_s + \omega L_u. \tag{10}$$

IV. Experiments and Results

A. Datasets

To evaluate the performance of the CLC^{2}SIE, extensive experiments were conducted on three public building extraction datasets, including the WHU aerial image dataset (WHU) [44], the CrowdAI mapping challenge dataset (CrowdAI) [45], and the building dataset of Typical Cities in China (TCC) [46].

1) WHU Aerial Building Dataset: The WHU Aerial Building Dataset sourced from Christchurch, New Zealand, encompasses an area of approximately 450 km^2. It comprises 8,188 images that capture buildings with diverse regional characteristics, sizes, and appearances. Each image features a resolution of 512 × 512 pixels at a spatial resolution of 0.3 m. The dataset has been partitioned into a training set consisting of 4736 images and a test set comprising 2416 images. Fig. 5(a) presents sample images from the dataset, where building instances are annotated with color-coded ground-truth labels.

2) CrowdAI Mapping Challenge Dataset: The CrowdAI mapping challenge dataset is a simplified version of the SpaceNet dataset. It includes more than 340 000 individual satellite images from multiple global cities, including Las Vegas, Paris, New York, and Shanghai. Each image has an RGB resolution of 300 × 300 pixels and a spatial resolution of 0.3 m. The dataset consists of a training set containing 280 741 images and a test set containing 60 697 images, providing annotations in MS-COCO format. Fig. 5(b) shows sample images from the CrowdAI dataset.

3) TCC Building Dataset: The TCC building dataset comprises 7260 image tiles and 63 886 building instances across four Chinese cities: Beijing, Shanghai, Shenzhen, and Wuhan. Among them, 5985 images are used for training, while the remaining 1275 images are used for test. Each image measures 500 × 500 pixels with a spatial resolution of 0.29 m. Notably, the dataset contains many densely distributed nonorthonormal buildings, which pose additional challenges for building instance extraction tasks. Fig. 5(c) displays sample images from the TCC dataset.

B. Experimental Settings

1) Dataset Settings: In the experiments, each dataset is divided into a training set, a validation set, and a test set. To evaluate the performance of the proposed method in semisupervised scenarios, 1%, 2%, 5%, 10%, and 20% samples from the training set form labeled subsets with different numbers of labels, and the remaining samples as unlabeled subsets. All experiments are performed with five data folds, and the final performance is calculated as the average of all five folds.

Different data augmentations are used in the training process, including the augmentation of labeled images, and the SDA and WDA of unlabeled images to achieve different perturbations. The details are shown in Table I.

2) Implementation Details: The basic networks of the teacher model and student model adopt the Mask R-CNN [27] with FPN [47], and use ImageNet pretrained ResNet-50 [48] as their backbone. All experiments are implemented in

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(a) (b) (c)

Fig. 5. Samples of the three datasets: (a) WHU. (b) CrowdAI. (c) TCC. First and third rows are the original images, and second and fourth rows are their ground-truth images.

TABLE I
DETAILS OF THE DATA AUGMENTATION USED IN OUR METHOD

Augmentation	Labeled data	SDA of unlabeled data	WDA of unlabeled data
Scale	limit∈(0.5, 1.5)	limit∈(0.5, 1.5)	limit∈(0.5, 1.5)
Rotate	p=0.25, angle∈(0,90)	p=0.25, angle∈(0,90)	p=0.25, angle∈(0,90)
Sharpness	p=0.25, limit∈(0, 1)	p=0.25, limit∈(0, 1)	-
Brightness	p=0.25, limit∈(0, 1)	p=0.25, limit∈(0, 1)	-
Gaussian blur	-	p=0.3, limit∈(0, 3)	-
Gaussian noise	-	p=0.3, limit∈(0, 12)	-
Invert	-	p=0.05	-

PyTorch [49] and MMDetection [50], and run on a NVIDIA RTX 3090 GPU.

The batch size for training is 5. The stochastic gradient descent (SGD) training [51] is employed and initialized with a learning rate of 0.0025. The weight decay and the momentum are set to 0.0001 and 0.9, respectively. Both models are trained for 180 000 iterations, in which the learning rate declined up to 120 000 iterations until 160 000 iterations.

C. Evaluation Metrics

In this study, the standard MS-COCO metrics [52] were employed for evaluating the method, including the mean average precision (mAP), and mean average recall (mAR) over multiple intersection over union (IoU) values. The mAP and mAR are computed at 10 IoU overlap thresholds ranging from 0.50 to 0.95 in 0.05 increments. The mAP and mAR are calculated as (10) and (11), respectively,

$$\text{mAP} = \frac{\text{AP}_{0.50} + \text{AP}_{0.55} + \cdots + \text{AP}_{0.90} + \text{AP}_{0.95}}{10} \tag{11}$$

$$\text{mAR} = \int_{0.5}^{1} \text{recall}(x)\text{dx} = \frac{2}{n}\sum_{i=1}^{n} \max\left(\text{IoU}\left(\text{gt}_i\right) - 0.5, 0\right). \tag{12}$$

D. Baselines

To investigate the effectiveness of CLC²SIE, six state-of-the-art methods are selected for comparison, including the full-supervised and semisupervised instance extraction methods. For the full-supervised methods, Mask R-CNN [27], RefineMask [53], and SOLOv2 [54] are served as baselines. Mask R-CNN is the basic network of the proposed method, and is widely used to extract building instances. RefineMask is an excellent two-stage instance segmentation method, which refines high-quality masks through incorporating fine-grained features during the instancewise segmentation process in a multistage manner. SOLOv2 is a typical representative of one-stage building instance extraction method, which can dynamically segment each instance without resorting to bounding box detection. The semisupervised methods include NB [33], PAIS [35], and SSBIE [17]. NB is the first semisupervised instance segmentation framework in computer vision field, which exploits more valuable information by further learning within boundary-relevant regions. PAIS adjusts the dynamic loss weight and achieves the excellent performance in semisupervised instance segmentation. SSBIE is the first SSL building instance extraction method, which adaptively acquires pseudo labels and optimizes the extracted buildings through multistage training.

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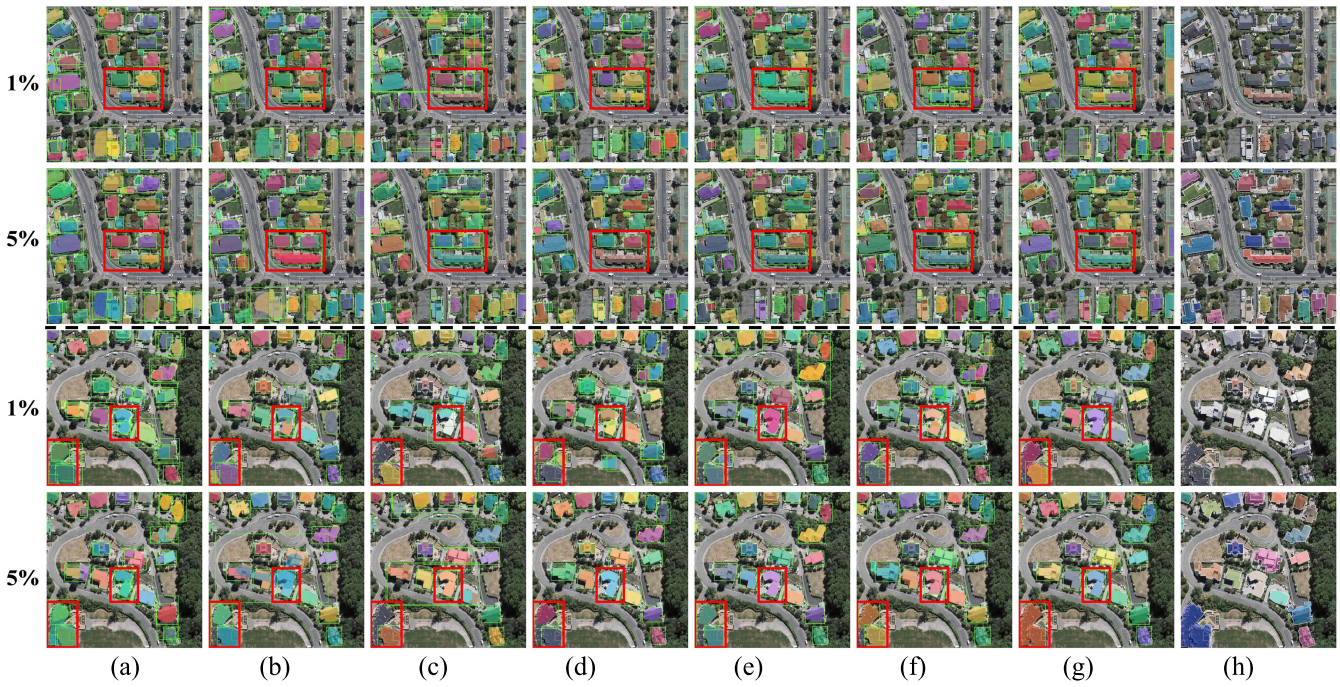


Fig. 6. Visualized results on the WHU dataset. (a) Mask R-CNN. (b) RefineMask. (c) SOLOv2. (d) NB. (e) PAIS. (f) SSBIE. (g) Our. (h) Image/GT.

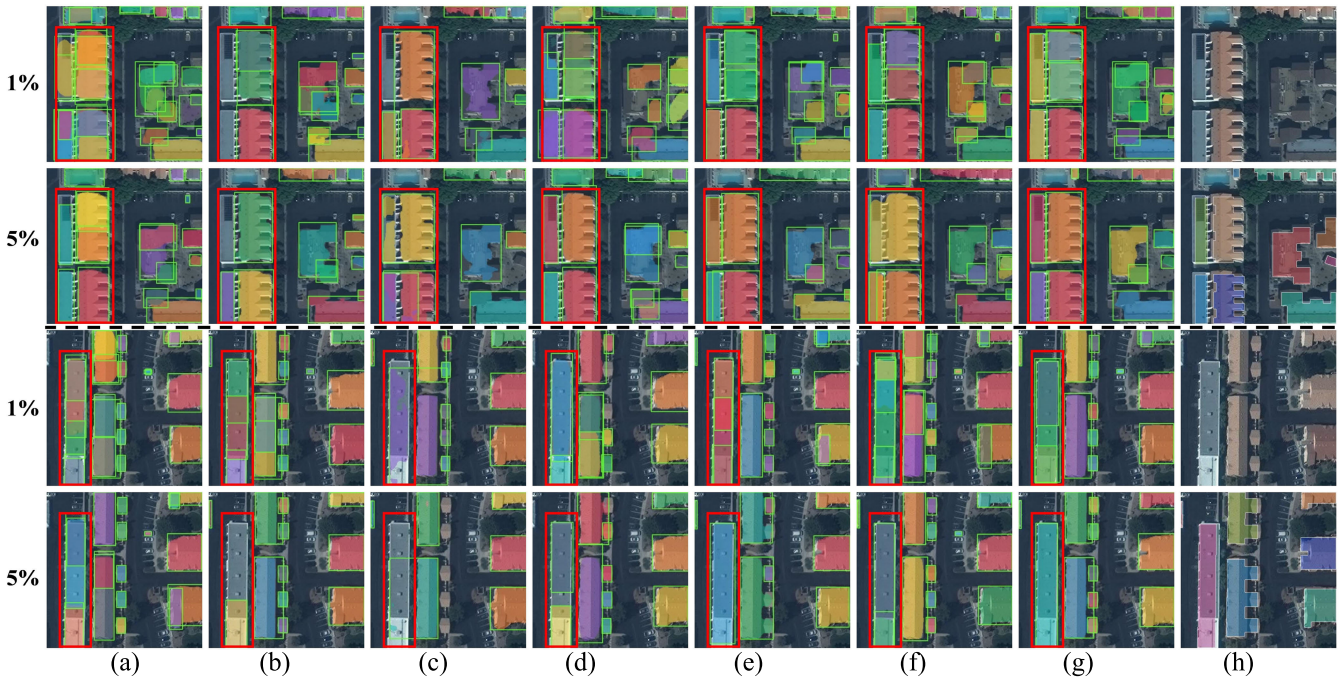


Fig. 7. Visualized results on the CrowdAI dataset. (a) Mask R-CNN. (b) RefineMask. (c) SOLOv2. (d) NB. (e) PAIS. (f) SSBIE. (g) Our. (h) Image/GT.

E. Overall Results

1) Results on the WHU Dataset: The experimental results on WHU dataset are listed in Table II. The proposed CLC^{2}SIE is superior to the baseline method. Compared with RefineMask, the best-performing fully supervised method, CLC^{2}SIE improves the mAP by 14.0%, 5.7%, 4.7%, 4.0%, and 2.3% on different ratios, respectively. Meanwhile, compared with the advanced semisupervised instance segmentation methods, CLC^{2}SIE also has obvious performance advantages. For example, when the label ratio is 1% and 5%, the mAP improves by 5.4% and 1.2% compared with SSBIE, respectively. The prominence of CLC^{2}SIE can be attributed to the effective utilization of background information and the high quality of pseudo labels. These results show that our method effectively enhances the performance of building instance extraction.

To further observe the result, two images representing challenging scenes from the WHU datasets are visualized in

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TABLE II
COMPARISON RESULTS ON THE WHU DATASETS

	Method	mAP					
		1%	2%	5%	10%	20%	100%
Full	Mask R-CNN	34.2%	47.1%	51.4%	56.3%	58.9%	63.5%
	RefineMask	39.3%	51.1%	54.7%	59.1%	61.9%	**65.3%**
	SOLOv2	20.3%	32.6%	45.1%	52.7%	56.6%	62.1%
Semi	NB	41.1%	51.6%	55.9%	60.8%	63.1%	-
	PAIS	34.9%	48.6%	56.7%	60.2%	64.4%	-
	SSBIE	47.9%	54.8%	58.2%	62.5%	**64.6%**	-
	CLC²SIE	**53.3%**	**56.8%**	**59.4%**	**63.1%**	64.2%	-

TABLE III
COMPARISON RESULTS ON THE CROWDAI DATASET

	Method	mAP					
		1%	2%	5%	10%	20%	100%
Full	Mask R-CNN	27.9%	38.9%	40.4%	46.6%	50.7%	56.2%
	RefineMask	32.3%	41.5%	48.2%	52.1%	54.6%	**58.5%**
	SOLOv2	18.2%	34.7%	41.2%	47.5%	50.8%	56.9%
Semi	NB	33.5%	40.6%	46.7%	51.1%	54.2%	-
	PAIS	44.5%	49.1%	51.9%	**53.8%**	54.9%	-
	SSBIE	36.3%	44.2%	50.6%	52.9%	55.1%	-
	CLC²SIE	**46.9%**	**50.0%**	**52.1%**	**53.8%**	**55.4%**	-

Fig. 6 with 1% and 5% labeled data as examples. As shown in the figure, the buildings in the proposed method demonstrate more integrity and exhibit more precise contours in diverse scenarios compared with the baselines. Specifically, as shown in the second row, our method can more accurately extract building instances with different shapes and densely distributed small buildings. In the scenery of large buildings with irregular visual characteristics, our method can more correctly distinguish the buildings compared to the baselines, as shown in the last row of Fig. 6. These visualization results further validate the outstanding performance of CLC²SIE.

2) Results on the CrowdAI Dataset: Table III reports the comparison of the proposed CLC²SIE with representative building instance extraction methods on the CrowdAI dataset. CLC²SIE achieves superior performance, with mAP of 46.9%, 50.0%, 52.1%, 53.8%, and 55.4% when the labeled ratio is 1%, 2%, 5%, 10%, and 20%, respectively. When fewer labeled samples are provided, CLC²SIE demonstrates significant improvement over Mask R-CNN, SOLOv2, RefineMask, NB, PAIS, and SSBIE. Specifically, at a labeling ratio of 1%, the mAP of CLC²SIE exceeded that of the aforementioned methods by 19.0%, 14.6%, 28.7%, 13.4%, 2.4%, and 10.6%, respectively. These results demonstrate that CLC²SIE has great potential in extracting building instances with insufficient training samples.

TABLE IV
COMPARISON RESULTS ON THE TCC DATASETS

	Method	mAP					
		1%	2%	5%	10%	20%	100%
Full	Mask R-CNN	3.4%	14.8%	21.5%	29.1%	36.6%	42.5%
	RefineMask	5.8%	18.0%	29.5%	36.2%	41.1%	**45.9%**
	SOLOv2	1.8%	9.9%	17.1%	28.9%	36.8%	42.0%
Semi	NB	6.4%	18.4%	29.8%	35.5%	40.2%	-
	PAIS	16.5%	28.3%	35.9%	38.2%	41.3%	-
	SSBIE	7.5%	20.5%	32.0%	37.2%	41.4%	-
	CLC²SIE	**21.7%**	**30.7%**	**37.1%**	**39.1%**	**42.3%**	-

Several samples from the CrowdAI dataset with 1% and 5% ratios are depicted in Fig. 7 to provide a more detailed observation of the results. The buildings surrounded by the red rectangles in Fig. 7(a)–(e) are not completely extracted, and some single buildings are mistakenly divided into two different instances. In contrast, the proposed CLC²SIE in Fig. 7(f) can more accurately segment building instances. Especially in the first scenario, our method effectively delineates the boundary region, even the buildings have complex contours. This success owes to the PLSC module guiding the model's perception of building boundaries during training. These results further demonstrate the superiority in building instance extraction.

3) Results on the TCC Dataset: As shown in Table IV, the values of mAP are lower than those of the WHU and CrowdAI datasets. This is because the TCC dataset contains many nonorthophoto images, which makes building instance extraction more challenging. Nevertheless, the proposed CLC²SIE performs best across all scenarios with different label ratio. Specifically, compared with PAIS, the second-best results, the performance has increased by 5.2%, 2.4%, 1.2%, 0.9%, and 1.0% on different ratios, respectively. Compared with best-performing RefineMask in the fully supervised methods, our method shows a larger performance gap, leading by 15.9% and 12.7% with only 1% and 5% labeled ratio, respectively. This promising performance shows that the proposed CLC²SIE is able to handle diverse challenging scenes.

The experimental results of the proposed method and baselines are visualized in Fig. 8. Although there are challenging scenarios with complex background interference in the TCC dataset, CLC²SIE extracts buildings with more complete structures and sharper contours. For instance, as presented in the second row of Fig. 8, the proposed method performs better than other methods in extracting building instances with clearer contours. In addition, CLC²SIE extracts more accurate buildings compared to PAIS, the method with best results in the baseline, as shown in the fourth row of Fig. 8. The visualized results further suggest that our method is promising.

V. DISCUSSION

A. Ablation Study

To investigate the effectiveness of the OLDC and PLSC modules, the ablation studies were analyzed on the WHU,

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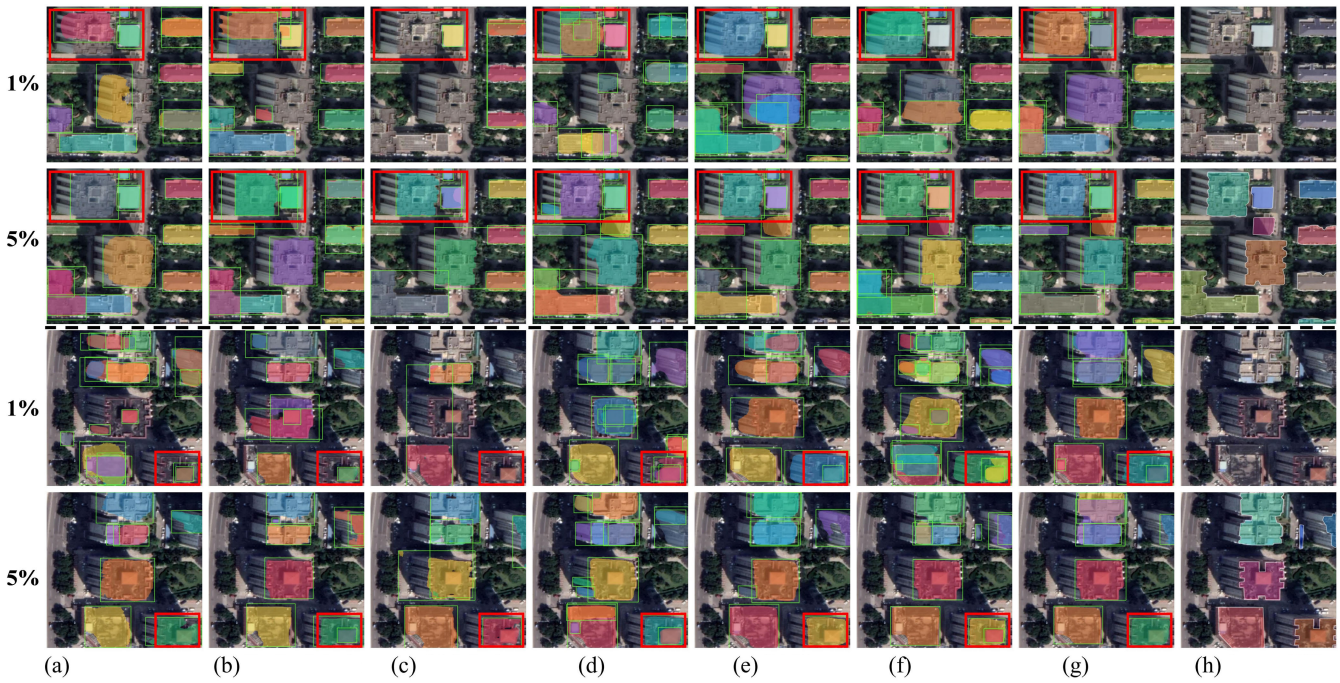


Fig. 8. Visualized results on the TCC dataset. (a) Mask R-CNN. (b) RefineMask. (c) SOLOv2. (d) NB. (e) PAIS. (f) SSBIE. (g) Our. (h) Image/GT.

CrowdAI, and TCC datasets with 5% label ratio. The experimental results are reported in Table V. Notably, removing OLDC module (w/o OLDC) indicates that the foreground boxes obtained by threshold filtering are calculated, ignoring the background candidates. Removing PLSC module (w/o PLSC) involves directly constraining the consistency of masks between the teacher and student models, without saliency guidance.

As shown in Table V, removing any module leads to a decline in performance. Specifically, CLC^{2}SIE performs best across all datasets, reaching 59.4%, 52.1%, and 37.1% in terms of mAP. After removing OLDC module, the mAP decreases by 2.3%, 2.6%, and 3.4% on the three datasets, respectively. We argue that the background candidate boxes hold potentially usable information, resulting in a better learning orientation of the student model. Furthermore, compared to our method, the mAP decreased by 1.4%, 1.9%, and 2.6% after removing the PLSC module, respectively. This indicates that the PLSC module reduces noises while improving the model extraction performance. After removing two modules, all metrics decrease substantially on three datasets. In particular, on the TCC dataset, the mAP, AP_{50}, and AP_{75} reached 30.3%, 54.0%, and 30.6%, respectively. The experimental results show that both OLDC and PLSC modules help to improve the performance of building instance extraction.

B. Pseudo Labels Results

The quality of pseudo labels determines the performance of building instance extraction. CLC^{2}SIE co-optimizes the model in an end-to-end manner to improve the quality of pseudo labels. To verify the quality of the pseudo labels, comparative experiment was conducted with a representative multistage approach, SSBIE [17], which uses the model pretrained by

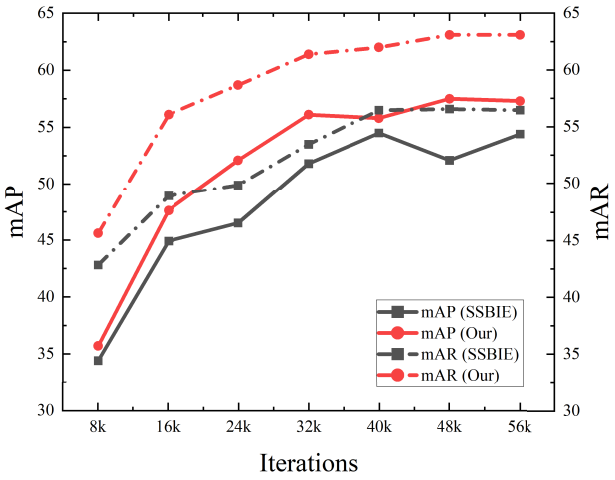


Fig. 9. Comparison results of pseudo labels.

label data to generate pseudo labels. Specifically, taking the WHU dataset with 5% label ratio as an example, the quality of pseudo labels of different method is verified by labeling in the original dataset. Both experiments use the same threshold of 0.8 for filtering. As shown in Fig. 9, the mAP and mAR value gradually increase as the number of iterations increments. SSBIE quickly reaches saturation during training, and mAP and mAR peak at 40k iterations. In contrast, the proposed CLC^{2}SIE outperforms the SSBIE. After 40k iterations, our method still ascends and obtains the best mAP. Meanwhile, the mAR is the highest, which means that more positive pseudo labels are generated. This can be interpreted as aggregating information of labeled data and unlabeled data in an end-to-end manner, the error accumulation between different modules is reduced by iterative co-optimization, enabling the model to achieve the global optimal performance [15].

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TABLE V
ABLATION STUDY RESULTS ON THREE DATASETS

Dataset	Method	mAP	AP_{50}	AP_{75}
WHU	CLC^{2}SIE	**59.4%**	**87.8%**	**70.2%**
	w/o OLDC	57.1% (2.3% ↓)	83.0% (4.8% ↓)	65.8% (4.4% ↓)
	w/o PLSC	58.0% (1.4% ↓)	86.9% (0.9% ↓)	68.1% (2.1% ↓)
	w/o OLDC, PLSC	55.5% (3.9% ↓)	82.8% (5.0% ↓)	62.9% (7.3% ↓)
CrowdAI	CLC^{2}SIE	**52.1%**	**82.4%**	**61.0%**
	w/o OLDC	49.5% (2.6% ↓)	79.2% (3.2% ↓)	57.4% (3.6% ↓)
	w/o PLSC	50.2% (1.9% ↓)	80.5% (1.9% ↓)	57.6% (3.4% ↓)
	w/o OLDC, PLSC	47.8% (4.3% ↓)	78.8% (3.6% ↓)	54.6% (6.4% ↓)
TCC	CLC^{2}SIE	**37.1%**	**63.2%**	**39.9%**
	w/o OLDC	33.7% (3.4% ↓)	58.9% (4.3% ↓)	35.3% (4.6% ↓)
	w/o PLSC	34.5% (2.6% ↓)	60.4% (2.8% ↓)	36.9% (3.0% ↓)
	w/o OLDC, PLSC	30.3% (6.8% ↓)	54.0% (9.2% ↓)	30.6% (9.3% ↓)



Fig. 10. Comparison results of pseudo labels. (a) Image. (b) GT. (c) SSBIE. (d) Our.

To investigate the pseudo labels of different methods, five samples from the WHU dataset are taken as an example for further observation in Fig. 10. As shown in Fig. 10(c), SSBIE ignores partial building instances and fails to generate complete pseudo labels for large buildings. This is mainly due to the fact that the multistage manner only uses limited labeled data to train the model, resulting in insufficient feature information and unable to accurately generate pseudo labels. Dissimilarly, Fig. 10(d) shows that buildings with different scales and types generate pseudo labels more completely. The experimental results suggest that the proposed method facilitates better quality of pseudo label and extraction of building instances.

TABLE VI
COMPARISON RESULTS OF DIFFERENT λ ON THE WHU DATASET

Indicator	mAP	AP_{50}	AP_{75}
0	57.1%	83.0%	65.8%
1	58.1%	86.9%	68.8%
Our	**59.4%**	**87.8%**	**70.2%**

C. Effect of λ

In Section III-B, the dynamic weight λ is used to control the weight of each background box in the loss L_u^{cls}. For comparison, several other indicators are examined, including 0 and 1. In Table VI, λ achieves the best values in all metrics. When λ is set to 1, the mAP decreases by 1.3% compared with our method. This phenomenon is due to the fact that candidates with low confidence in the background box cannot be suppressed, which misleads model learning. λ set to 0 indicates that the background box does not contribute to L_u^{cls}, at which the mAP drops to 57.1%. This means that the model loses useful buildings information from the background during training, harming the performance. These results suggest that λ effectively contributes to the improvement of building instance extraction from HR RSIs.

To analyze the impact of weight λ on L_u^{cls}, the features of different λ are visualized to observe the concerns of model learning. As shown in Fig. 11, the red rectangles represent the

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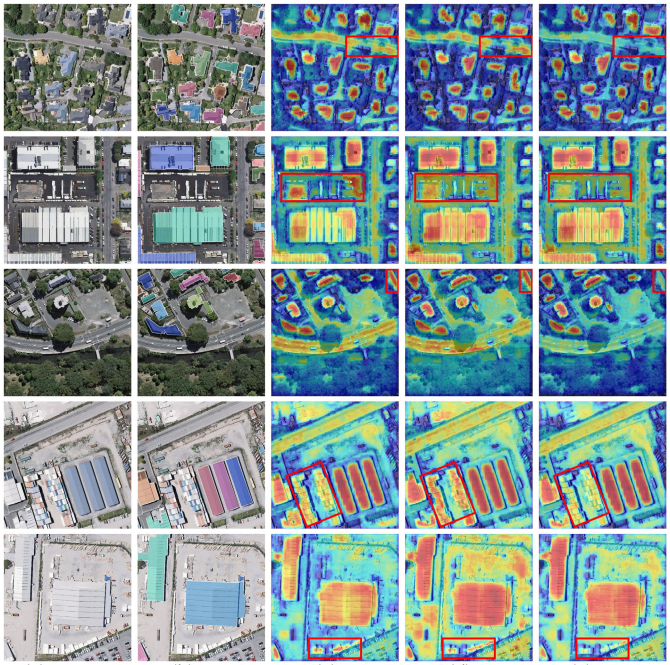


Fig. 11. Visualization of features in different weights λ. (a) Image. (b) GT. (c) 0. (d) 1. (e) Our.

TABLE VII
COMPARISON RESULTS OF EDGE DETECTION OPERATORS ON THE WHU DATASET

Method	mAP	AP_{50}	AP_{75}
Sobel	58.6%	86.9%	69.2%
Canny	58.0%	86.1%	68.4%
Laplace	**59.4%**	**87.8%**	**70.2%**

background interference to focus. When 0 is employed, the buildings in the third column of Fig. 11 manifest as light red, and the background part has an incorrect focus, especially in the second and fifth scenes. This occurs because background information is disregarded, and it is challenging for the model to distinguish between the foreground and the background. When λ is set to 1, the feature values in the building area become higher, which means that the addition of background makes the model pay more attention to the building features. Nevertheless, nonbuildings in the background that are not properly suppressed will introduce redundant interference, misleading model's attention. Dissimilarly, the buildings in the fifth column exhibit higher feature values and lower background values. This indicates that our approach effectively suppresses background interference while concentrating on the buildings.

D. Effect of Edge Detection Operators

The PLSC module adopts the Laplace operator to extract saliency boundaries. To investigate the effect of the operator, additional experiments are conducted with two substitutable operators. Specifically, taking the WHU dataset with 5% label ratio as an example, the performance of several edge detection operators is discussed, including Laplace [43], Sobel [55], and Canny [56]. As shown in Table VII, the highest mAP is achieved by applying the Laplace operator, reaching 59.4%, which is higher than 0.8% and 1.4% using the Sobel and Canny operators, respectively. This may be due to its isotropic nature and ability to detect edges in all orientations without directional bias, which is particularly advantageous for extracting building boundaries in complex urban scenes [57], [58].

TABLE VIII
COMPARISON RESULTS OF DIFFERENT SAMPLING RATIOS ON THE WHU DATASET

Ratio	mAP	AP_{50}	AP_{75}
1: 1	58.7%	87.5%	69.3%
2: 1	58.7%	87.3%	69.1%
4: 1	57.9%	86.9%	68.8%
1: 2	59.2%	87.6%	**70.3%**
1: 4	**59.4%**	**87.8%**	70.2%

E. Parameter Sensitivity Analysis

Two hyperparameters, the sampling ratio (see Section III-A) and the foreground threshold in the OLDC module (see Section III-B), have the potential to impact the model's performance. The sensitivity analysis of these hyperparameters is conducted in this section.

1) Effect of Sampling Ratio: In Section III-A, labeled and unlabeled RSIs are randomly sampled by the data sampling ratio, and then fed into the subsequent network to learn together. To explore the impact of sampling ratio, the experiments were conducted with different samples on the WHU dataset with 5% label ratio. The results are reported in Table VIII.

The AP values gradually increase as the number of unlabeled data increments. We argue that abundant unlabeled data allows the model to learn building features more generically. In this phase, increasing the amount of unlabeled data will improve the extraction accuracy. When the ratio is set to 1:4, the mAP value is optimal. Nevertheless, when the ratio of labeled data increases, the mAP value decreases instead. This can be explained that the labeled data establishes the basis for model learning in SSL, which is further enhanced through the incorporation of unlabeled data [59]. Therefore, when the unlabeled data is scarce, it is prone to make certain erroneous judgments on unknown data and generate inaccurate pseudo labels, which impedes the achievement of high accuracy. In addition, there is a risk of overfitting when the model learning is too biased toward labeled data, which may lead to poor generalization [18].

2) Effect of ω: In Section III-D, ω is used to weight unsupervised loss L_u. To validate the effectiveness of weight ω, the experiments were conducted with various weights on the WHU dataset with 5% label ratio. The results are presented in Table IX. To be specific, as ω increases from 0 to 1.0,

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TABLE IX
COMPARISON RESULTS OF ω ON THE WHU DATASET

Indicator	mAP	AP_{50}	AP_{75}
0	52.1%	81.5%	63.0%
0.5	57.8%	86.7%	69.1%
1.0	**59.4%**	**87.8%**	**70.2%**
1.5	59.2%	87.7%	69.9%

TABLE X
COMPARISON RESULTS OF DIFFERENT FOREGROUND THRESHOLDS ON THE WHU DATASET

Threshold	mAP	AP_{50}	AP_{75}
0.5	57.8%	87.0%	68.2%
0.6	58.2%	87.1%	69.5%
0.7	58.9%	87.3%	69.7%
0.8	**59.4%**	**87.8%**	**70.2%**
0.9	59.3%	**87.8%**	70.0%

there is a general trend of accuracy improvement, reaching its peak at 59.4% when ω is set to 1.0. This suggests that the incorporation of unsupervised loss introduces additional valuable information from unlabeled data, thereby enhancing model training. However, as the weight ω continues to rise, the model's performance slightly declines, dropping to 59.2% when the threshold is set to 1.5. This decrease may be attributed to the fact that an excessively high weight on the local loss causes the model to pay excessive attention to unlabeled data, potentially leading to the neglect of global information [19].

3) Effect of Foreground Threshold: The foreground threshold aims to divide the candidate boxes generated by the student model into foreground or background. The values of threshold may affect the performance of CLC²SIE. In the section, experiments are performed on the WHU dataset with five values, ranging from 0.5 to 0.9 with a step size of 0.1, to investigate the effect of foreground threshold. The experimental results are illustrated in Table X. The AP values exhibit an increasing trend as the threshold increases gradually. When the threshold is 0.8, CLC²SIE performs best. This can be explained by (1), high threshold means that candidate boxes with high confidence are endorsed to weight loss, which determines the orientation of model learning.

The precision–recall curves generated from different thresholds are shown in Fig. 12. The red curve corresponding to the 0.8 threshold lies above other thresholds, indicating a more favorable trade-off between precision and recall. These results suggest that 0.8 threshold is more appropriate for extracting building instances and exhibits strong robustness and good stability.

F. Model Efficiency

To examine the efficiency of the proposed method, the comparison experiments with six baseline methods are performed on the WHU dataset. Fig. 13 reports the FPS and model

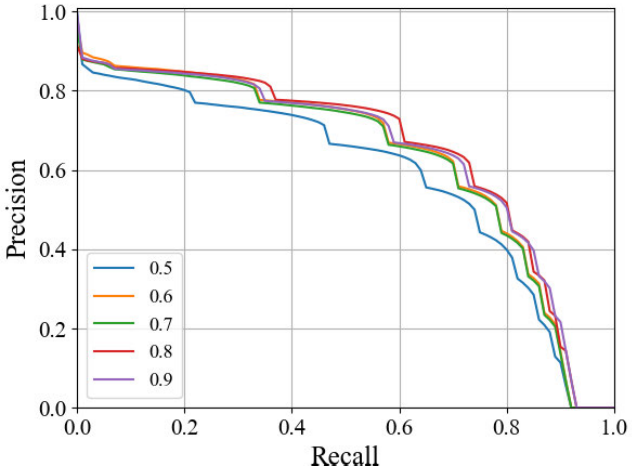


Fig. 12. Precision–recall curves of different thresholds.

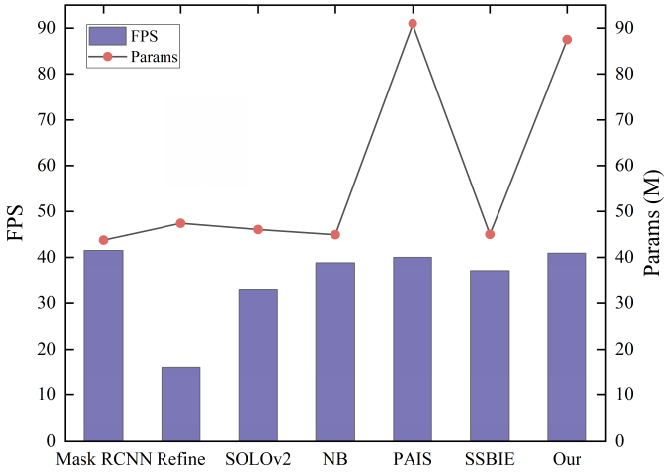


Fig. 13. Comparison results of model FPS and parameters.

parameter. Compared with the fully supervised and semisupervised baseline methods, CLC²SIE achieves the highest FPS. This is because the proposed method only uses the trained teacher model or student model for inference. Compared with the advanced PAIS of SSIS, CLC²SIE possesses parameter advantages while attaining superior performance. Although our method has almost twice the number of parameters compared to Mask R-CNN, it improves the model performance in the limited labeled data scenario. We argue that the proposed method offers a great trade-off between accuracy and efficiency improvement.

VI. CONCLUSION

In this study, an end-to-end semisupervised method, called CLC²SIE, is proposed to extract building instances from HR RSIs. The OLDC module dynamically converts valuable background information and enhances the model's perception of buildings in complex scenes. The PLSC module improves model's awareness of building contours by saliency guidance. By properly training these modules jointly in end-to-end manner, CLC²SIE simultaneously improves the quality of the pseudo label and the performance of building instance extraction. This method has been evaluated on the WHU,

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CrowdAI, and TCC building dataset. The experimental results show that CLC^{2}SIE outperforms the baseline methods. To our knowledge, dynamic converting building features from complex background in RSIs is novel in SSL building instance extraction. This technique is expected to serve as a methodology reference for various SSL applications on RSIs.

However, the real-scene RSIs often suffer from extreme sample distribution imbalances and significant buildings stylistic differences between labeled and unlabeled data. This poses greater challenges to SSL approaches. In the future work, we will further explore domain adaptive method to address these challenges in practical applications.

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